

CSE 5830: Probabilistic Graph Analysis

Homework 3

Due 11:59PM, Wednesday 9/30/2026

Show all your work to get a full score.

Problem 1: Noisy-AND (15 points)

A Noisy-AND model (independence of the causal mechanisms when all X s are 1) with binary random variables and leaky parameters $(\lambda_0, \lambda_1, \dots, \lambda_K)$.

(a) Expression for $P(Y = 0 \mid X_1, \dots, X_K)$ (10 points)

Setup / convention. Noisy-AND is the De Morgan dual of noisy-OR. In a deterministic AND gate, a single input $X_i = 0$ is enough to force $Y = 0$, and $Y = 1$ happens in only one way: every input equals 1.

X_1	X_2	$Y = X_1 \wedge X_2$
0	0	0
0	1	0
1	0	0
1	1	1

For each parent, introduce an independent mechanism variable Z_i , where $Z_i = 1$ means “ X_i forces $Y = 0$ ”. An input that is present can never force an AND gate to 0, so

$$P(Z_i = 1 \mid X_i = 1) = 0, \quad P(Z_i = 1 \mid X_i = 0) = \lambda_i.$$

That is, λ_i is the probability that an absent input ($X_i = 0$) succeeds in forcing $Y = 0$. The leak Z_0 is always active, with $P(Z_0 = 1) = \lambda_0$: the probability that $Y = 0$ even when every input is 1. Then $Y = 1$ if and only if $Z_0 = Z_1 = \dots = Z_K = 0$.

Derivation. Since $Y = 1$ requires that no mechanism fires, and the mechanisms are independent (independence of the causal mechanisms),

$$P(Y = 1 \mid X_1, \dots, X_K) = P(Z_0 = 0) \prod_{i=1}^K P(Z_i = 0 \mid X_i).$$

Each factor is

$$P(Z_i = 0 \mid X_i) = \begin{cases} 1, & X_i = 1, \\ 1 - \lambda_i, & X_i = 0, \end{cases}$$

so parents with $X_i = 1$ contribute a factor of 1 and drop out. Hence

$$P(Y = 1 \mid X_1, \dots, X_K) = (1 - \lambda_0) \prod_{i: X_i=0} (1 - \lambda_i),$$

and taking the complement,

$$P(Y = 0 \mid X_1, \dots, X_K) = 1 - (1 - \lambda_0) \prod_{i: X_i=0} (1 - \lambda_i)$$

Sanity check. With no leak ($\lambda_0 = 0$) and $\lambda_i = 1$ for all i : if every $X_i = 1$, the product is empty (equal to 1), so $P(Y = 0) = 0$; if any $X_i = 0$, the product contains a factor $1 - 1 = 0$, so $P(Y = 0) = 1$. This is exactly the deterministic AND gate.

(b) Numerical case (5 points)

$K = 4$, $\lambda_0 = 0$, $\lambda_1 = 0.9$, $\lambda_2 = 0.75$, $\lambda_3 = 0.8$, $\lambda_4 = 0.85$.

Only X_2 and X_3 are off, so only λ_2 and λ_3 enter the product; $X_1 = X_4 = 1$ contribute factors of 1. With no leak, $1 - \lambda_0 = 1$. Using the formula from (a):

$$\begin{aligned} P(Y = 0 \mid X_1 = 1, X_2 = 0, X_3 = 0, X_4 = 1) &= 1 - (1 - \lambda_0)(1 - \lambda_2)(1 - \lambda_3) \\ &= 1 - (1 - 0)(1 - 0.75)(1 - 0.8) \\ &= 1 - (1)(0.25)(0.2) \\ &= 1 - 0.05 \\ &= \boxed{0.95} \end{aligned}$$

Sanity check. Two inputs are off, and each has a high chance (0.75 and 0.8) of forcing $Y = 0$. So $Y = 1$ requires both of those mechanisms to fail, which happens with probability only $0.25 \times 0.2 = 0.05$. A value of $P(Y = 0)$ close to, but below, 1 is what we expect.

Problem 2: Factor product (15 points)

Given factors:

X	Y	$\phi_1(X, Y)$	Y	Z	$\phi_2(Y, Z)$
1	1	0.8	1	1	0.2
1	2	0.5	1	2	0.2
2	1	0.5	2	1	0.9
2	2	0.6	2	2	1.0

(a) $\psi(X, Y, Z) = \phi_1(X, Y) \phi_2(Y, Z)$ (10 points)

X	Y	Z	$\phi_1(X, Y) \cdot \phi_2(Y, Z)$	$\psi(X, Y, Z)$
1	1	1	$0.8 \cdot 0.2$	0.16
1	1	2	$0.8 \cdot 0.2$	0.16
1	2	1	$0.5 \cdot 0.9$	0.45
1	2	2	$0.5 \cdot 1.0$	0.5
2	1	1	$0.5 \cdot 0.2$	0.1
2	1	2	$0.5 \cdot 0.2$	0.1
2	2	1	$0.6 \cdot 0.9$	0.54
2	2	2	$0.6 \cdot 1.0$	0.6

(b) $\psi(X, Z \mid Y = 1)$ (5 points)

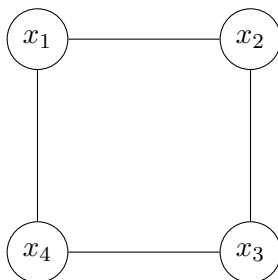
This is a factor reduction to the context $Y = 1$: keep only the rows of $\psi(X, Y, Z)$ from (a) with $Y = 1$ and drop the Y column. Reduction alone does not normalize, so the unnormalized values are shown first. To read $\psi(X, Z \mid Y = 1)$ as a conditional distribution, divide each entry by their sum, $0.16 + 0.16 + 0.1 + 0.1 = 0.52$.

X	Z	$\psi[Y = 1]$ (reduced)	$\psi(X, Z \mid Y = 1)$ (normalized)
1	1	0.16	$0.16/0.52 \approx 0.308$
1	2	0.16	$0.16/0.52 \approx 0.308$
2	1	0.1	$0.1/0.52 \approx 0.192$
2	2	0.1	$0.1/0.52 \approx 0.192$
Sum		0.52	1

Problem 3: Pairwise Markov network (10 points)

$$p(\mathbf{x}) \propto \phi(x_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x_1)$$

(a) Draw the network (2 points)



(b) Conditionals in terms of ϕ (8 points)

General method. Write the conditional as a ratio of the joint to its sum over the queried variable. The partition function Z cancels, and any factor that does not contain the queried variable takes the same value in the numerator and in every term of the denominator, so it cancels too. Only the factors touching the queried variable (that is, the edges to its two neighbors) remain.

$p(x_1 \mid x_2, x_4)$.

$$p(x_1 \mid x_2, x_4) = \frac{\frac{1}{Z} \phi(x_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x_1)}{\sum_{x'_1} \frac{1}{Z} \phi(x'_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x'_1)}.$$

The factors $\phi(x_2, x_3)$ and $\phi(x_3, x_4)$ do not contain x_1 , so they cancel along with $\frac{1}{Z}$:

$$p(x_1 \mid x_2, x_4) = \frac{\phi(x_1, x_2) \phi(x_4, x_1)}{\sum_{x'_1} \phi(x'_1, x_2) \phi(x_4, x'_1)}.$$

$p(x_2 \mid x_1, x_3)$. The factors containing x_2 are $\phi(x_1, x_2)$ and $\phi(x_2, x_3)$; the factors $\phi(x_3, x_4)$ and $\phi(x_4, x_1)$ cancel.

$$p(x_2 \mid x_1, x_3) = \frac{\phi(x_1, x_2) \phi(x_2, x_3)}{\sum_{x'_2} \phi(x_1, x'_2) \phi(x'_2, x_3)}.$$

$p(x_3 \mid x_2, x_4)$. The factors containing x_3 are $\phi(x_2, x_3)$ and $\phi(x_3, x_4)$; the factors $\phi(x_1, x_2)$ and $\phi(x_4, x_1)$ cancel.

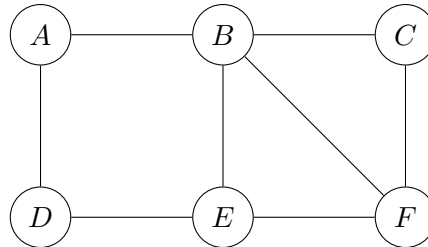
$$p(x_3 \mid x_2, x_4) = \frac{\phi(x_2, x_3) \phi(x_3, x_4)}{\sum_{x'_3} \phi(x_2, x'_3) \phi(x'_3, x_4)}.$$

$p(x_4 \mid x_1, x_3)$. The factors containing x_4 are $\phi(x_3, x_4)$ and $\phi(x_4, x_1)$; the factors $\phi(x_1, x_2)$ and $\phi(x_2, x_3)$ cancel.

$$p(x_4 \mid x_1, x_3) = \frac{\phi(x_3, x_4) \phi(x_4, x_1)}{\sum_{x'_4} \phi(x_3, x'_4) \phi(x'_4, x_1)}.$$

Observation. Each conditional depends only on the variable's two neighbors in the cycle. This is the local Markov property: a node is independent of the rest of the network given its neighbors.

Problem 4: Markov network structure (20 points)



(a) All factors that factorize over the graph (6 points)

Edges: $A-B$, $A-D$, $B-C$, $B-E$, $B-F$, $C-F$, $D-E$, $E-F$ (8 edges).

Cliques (every pair of nodes in the set must be joined by an edge):

- Single nodes: $\{A\}, \{B\}, \{C\}, \{D\}, \{E\}, \{F\}$.
- Edges: the 8 edges listed above.
- Triangles: $\{B, E, F\}$ (edges $B-E$, $E-F$, $B-F$) and $\{B, C, F\}$ (edges $B-C$, $C-F$, $B-F$).
- No 4-cliques: e.g. $\{A, B, D, E\}$ is missing $A-E$ and $B-D$, and $\{B, C, E, F\}$ is missing $C-E$.

Maximal cliques: the two triangles $\{B, E, F\}$ and $\{B, C, F\}$, plus the edges not contained in either triangle: $\{A, B\}$, $\{A, D\}$, $\{D, E\}$.

A distribution factorizes over this graph if it can be written as a product of factors over its maximal cliques (factors over smaller cliques can be absorbed into these):

$$P(A, B, C, D, E, F) = \frac{1}{Z} \phi_1(A, B) \phi_2(A, D) \phi_3(D, E) \phi_4(B, E, F) \phi_5(B, C, F),$$

$$Z = \sum_{a,b,c,d,e,f} \phi_1(a, b) \phi_2(a, d) \phi_3(d, e) \phi_4(b, e, f) \phi_5(b, c, f).$$

(b) Gibbs distribution as a pairwise MN (6 points)

A pairwise Markov network has one factor per edge (and optionally one per node), even for edges that lie inside a triangle. Using the 8 edges:

$$P(A, B, C, D, E, F) = \frac{1}{Z} \phi(A, B) \phi(A, D) \phi(B, C) \phi(B, E) \phi(B, F) \phi(C, F) \phi(D, E) \phi(E, F),$$

$$Z = \sum_{a,b,c,d,e,f} \phi(a, b) \phi(a, d) \phi(b, c) \phi(b, e) \phi(b, f) \phi(c, f) \phi(d, e) \phi(e, f).$$

Unlike in (a), the triangles $\{B, E, F\}$ and $\{B, C, F\}$ are not given their own 3-variable factors; each is represented only through its three pairwise edge factors.

(c) Local Markov independencies (6 points)

In a Markov network, each node is independent of all other nodes given its neighbors (its Markov blanket): $X \perp (\mathcal{X} \setminus (\{X\} \cup \text{MB}(X))) \mid \text{MB}(X)$.

Node	Neighbors (Markov blanket)	Local independence
A	$\{B, D\}$	$A \perp \{C, E, F\} \mid \{B, D\}$
B	$\{A, C, E, F\}$	$B \perp \{D\} \mid \{A, C, E, F\}$
C	$\{B, F\}$	$C \perp \{A, D, E\} \mid \{B, F\}$
D	$\{A, E\}$	$D \perp \{B, C, F\} \mid \{A, E\}$
E	$\{B, D, F\}$	$E \perp \{A, C\} \mid \{B, D, F\}$
F	$\{B, C, E\}$	$F \perp \{A, D\} \mid \{B, C, E\}$

(d) Is $A \perp F \mid \{D, C\}$? (2 points)

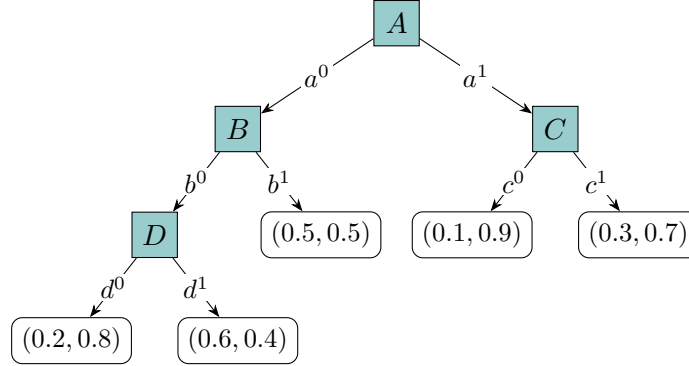
No. In a Markov network, $A \perp F \mid \{D, C\}$ holds only if every path from A to F passes through an observed node in $\{D, C\}$ (separation). The path

$$A - B - F$$

uses only the unobserved node B , so it is active. ($A - B - E - F$ is another active path.) Hence A and F are not separated by $\{D, C\}$, and they are not independent given D and C . Observing D and C blocks only the paths $A - D - E - F$ and $A - B - C - F$, not the route through B .

Problem 5: Tree-structured CPD (10 points)

Leaves list $(P(e^0), P(e^1))$.



To read the tree CPD, start at the root and follow the branch matching each observed value until a leaf is reached. Variables not on that path do not affect the result.

(a) $P(E = e^0 \mid a^0, b^1, c^1, d^1)$

Path: $A \xrightarrow{a^0} B \xrightarrow{b^1} (0.5, 0.5)$. The values of C and D are irrelevant in this context.

$$P(E = e^0 \mid a^0, b^1, c^1, d^1) = \boxed{0.5}$$

(b) $P(E = e^1 \mid a^1, b^1, c^1, d^1)$

Path: $A \xrightarrow{a^1} C \xrightarrow{c^1} (0.3, 0.7)$. The values of B and D are irrelevant in this context.

$$P(E = e^1 \mid a^1, b^1, c^1, d^1) = \boxed{0.7}$$

(c) $P(E = e^1 \mid a^0, b^0, c^0, d^0)$

Path: $A \xrightarrow{a^0} B \xrightarrow{b^0} D \xrightarrow{d^0} (0.2, 0.8)$. The value of C is irrelevant in this context.

$$P(E = e^1 \mid a^0, b^0, c^0, d^0) = \boxed{0.8}$$

(d) **Is** $E \perp_c B \mid a^0$?

No. In the context a^0 , the tree splits on B , and the two branches lead to different distributions over E :

$$P(E \mid a^0, b^1) = (0.5, 0.5), \quad P(E \mid a^0, b^0, d^0) = (0.2, 0.8), \quad P(E \mid a^0, b^0, d^1) = (0.6, 0.4).$$

For example, with d^0 , changing B from b^1 to b^0 changes $P(e^1)$ from 0.5 to 0.8. So $P(E \mid a^0, B, \cdot)$ depends on B , and E is not context-specifically independent of B given a^0 .

(e) Is $E \perp_c C \mid d^1$?

No. The context d^1 fixes only D , and D appears only in the a^0, b^0 subtree. It does not affect the a^1 branch, where the tree splits on C :

$$P(E \mid a^1, c^0, d^1) = (0.1, 0.9), \quad P(E \mid a^1, c^1, d^1) = (0.3, 0.7).$$

So with $A = a^1$, changing C changes the distribution of E even when d^1 is fixed. (Under a^0 , C is irrelevant, but context-specific independence must hold for every value of the remaining variables, and it fails when $A = a^1$.) Hence E is not context-specifically independent of C given d^1 .

Use of Ai

In this assignment, I used Ai to help me understand the assignment give me hints for each question as well as helping whenever I got stuck in a question. Also used AI to write and debug my tex code. However, I solved all the question of the assignment on my own.